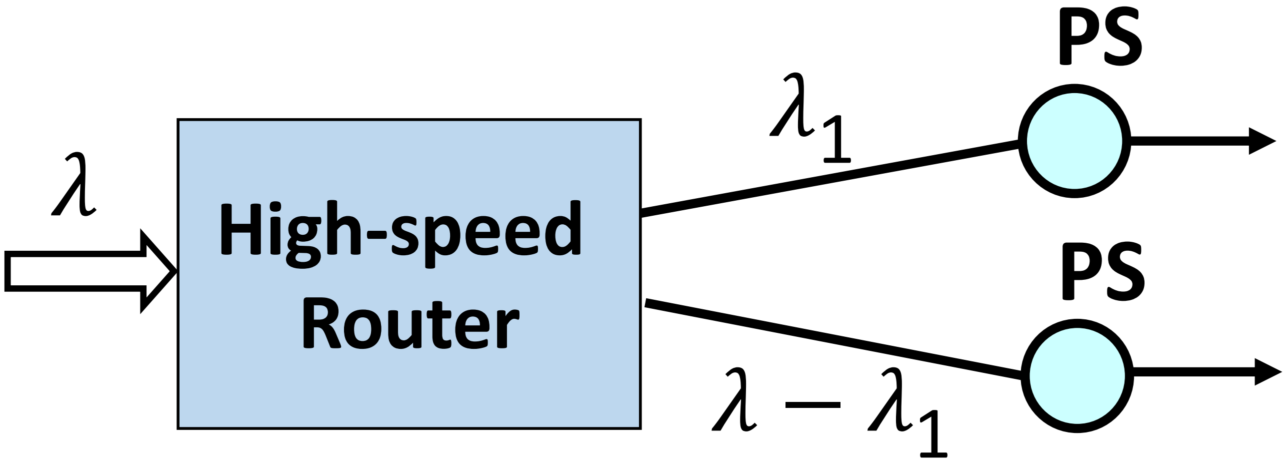
Processing speed μ_1 is a function of power allocation $\mu_1(P_1)$ where

Power allocated to server 1 = P_1



Processing speed $\mu_2(P_{total}-P_1)$

Processing speed

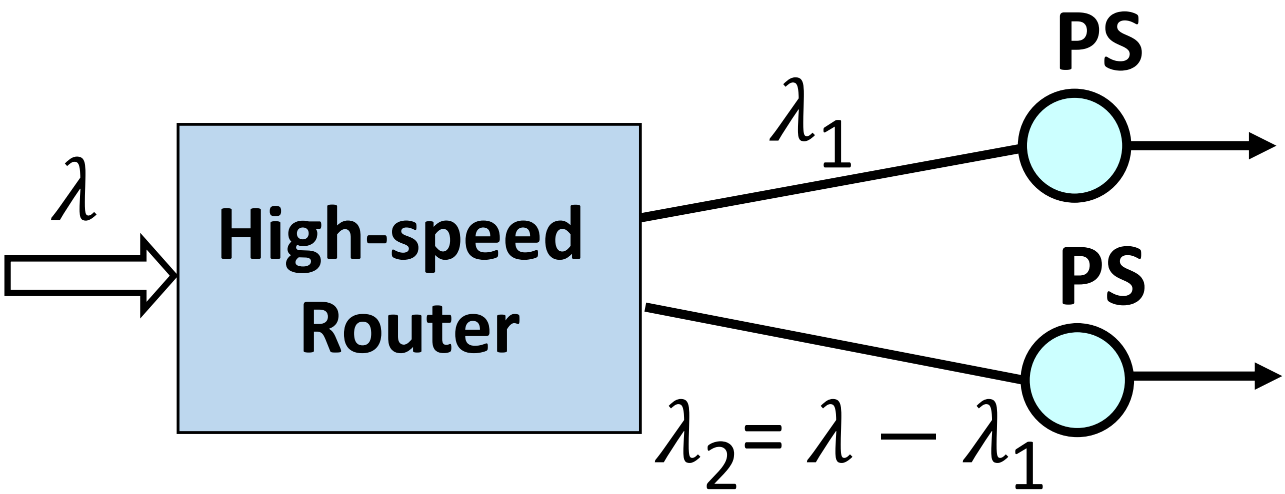
= $s_b + \boldsymbol{\alpha}$ (allocated power – b)

For constants s_b and **b**

$$\mu_1(P_1) = s_b + \boldsymbol{\alpha}\,(P_1 - b)$$
$$\mu_2(P_{total}-P_1) = s_b + \boldsymbol{\alpha}\,(P_{total} - P_1 - b)$$

Processing speed μ_1 is a function of power allocation $\mu_1(P_1)$ where

Power allocated to server 1 = P_1



Processing speed $\mu_2(P_{total}\text{-}P_1)$

Mean response time =

$$\frac{\lambda_1}{\lambda}\frac{1}{\mu_1 - \lambda_1} + \frac{\lambda_2}{\lambda}\frac{1}{\mu_2 - \lambda_2}$$

Two degrees of freedom: λ_1 and P_1

For a given P_1 :

Search for the $\lambda 1$ that minimizes mean response time